

WHO Global Air Quality Guidelines: Particulate Matter and Ozone Health Assessment

Page 1 of 3 | Document ID: doc_en_03

Executive Summary: Objectives and Scope of the Global Guidelines

The World Health Organization (WHO) provides evidence-based air quality guideline levels to protect public health worldwide.

The guidelines synthesize epidemiological evidence linking ambient air pollution exposure to mortality and morbidity endpoints.

Air pollution is identified as one of the largest global environmental health risks, contributing to over 7 million premature deaths.

Particulate matter (PM_{2.5} and PM₁₀) and tropospheric ozone represent the primary pollutants causing cardiopulmonary diseases.

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Page 2 of 3 | Document ID: doc_en_03

Quantitative Air Quality Guideline Thresholds for Particulate Matter

The recommended annual mean guideline level for fine particulate matter PM_{2.5} is established at 5 micrograms per cubic meter.

The 24-hour mean guideline level for PM_{2.5} is set at 15 µg/m³, not to be exceeded more than 3 to 4 days per calendar year.

For coarse particles (PM₁₀), the recommended annual guideline concentration is 15 µg/m³ and the 24-hour threshold is 45 µg/m³.

Nitrogen dioxide (NO₂) guidelines stipulate an annual mean concentration of 10 µg/m³ and a 24-hour concentration limit of 25 µg/m³.

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Page 3 of 3 | Document ID: doc_en_03

Ozone Guideline Levels and Recommended Public Health Interventions

The peak season 8-hour daily maximum ozone (O₃) concentration guideline is set at 60 µg/m³ to prevent respiratory hospital admissions.

Exposure above guideline thresholds is quantitatively correlated with increased risks of stroke, ischemic heart disease, and lung cancer.

WHO urges member states to enact clean energy transitions, phase out coal combustion, and expand zero-emission public transport.

Targeted air quality alert systems should advise vulnerable populations, including asthmatic children, during high pollution episodes.